Lotka-Volterra model

Computational simulation

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ELTE

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TUDOMÁNYEGYETEM

1 Motivation

The Lotka-Volterra model is a combination of differential equations that describe the interaction between two (or as we will see later even three or more) populations. We will see that while nature itself looks complicated, taking only a part out of it and trying to simulate that portion is not that hard. This model was independently proposed by two scientist, one statistician (Alfred J. Lotka) and a mathematician (Vito Volterra). In this report I will try to show how multiple species can live together in harmony or live by preying on each other.

2 Introduction and theoretical background

The Lotka-Volterra model in general tries to describe the time evolution of the number of a species. It does that by factoring in natural growth rate, death rate and the effect other species have on the given population. The results vary between periodic solutions, extinction, stabilisation of their specimen number and sometimes overpopulation. To stop the latter from happening one can introduce a factor k that represents the available resources. When the results are periodic it happens like this: let's say there are 2 species in a simulation, one is the prey and the other one is a carnivore. First there is a lot of prey, and few carnivores. Because there is a lot of food to find, the carnivores are multiplying fast, which results in less prey surviving. With fewer prey, many of the predators starve to death. With less risk of being eaten the preys' number will skyrocket, and the cycle continues.

For two species the formula looks like this:

$$\begin{aligned}\frac{dx}{dt} &= \alpha x - \beta xy,\\ \frac{dy}{dt} &= \delta xy - \gamma y, \end{aligned} \tag{1}$$

where $\frac{dx}{dt}$ and $\frac{dy}{dt}$ are the time evolution of specimen x and y. α is the growth factor of the prey, while $-\gamma y$ the death factor of the predator. $-\beta xy$ and δxy represent their effect on each other with corresponding signs whether their interaction is beneficial to them or not. Of course these are only a fraction of the variables their number depends on, and an even smaller fraction of the whole ecosystem.

Adding a third species in z will result in the following equations:

$$\begin{aligned}\frac{dx}{dt} &= \alpha x - \beta xy,\\ \frac{dy}{dt} &= -cy + \delta xy - eyz,\\ \frac{dz}{dt} &= -fz + gyz. \end{aligned} \tag{2}$$

The natural birth/death rates are αx, cy and fz. Their interactions with each other are $-\beta xy$, δxy, eyz, gyz. The problem with this model is that it only allows for a chain of predation, where only one species interacts with the other two. Generalizing this model will yield the following model:

$$P(T) = \begin{bmatrix} p_1(t) \\ p_2(t) \\ p_3(t) \end{bmatrix}, \tag{3}$$

$$D = \begin{bmatrix} 0 & d_{1,2} & d_{1,3} \\ d_{2,1} & 0 & d_{2,3} \\ d_{3,1} & d_{3,2} & 0 \end{bmatrix}, \tag{4}$$

$$G = \begin{bmatrix} g_1 \\ g_2 \\ g_3 \end{bmatrix}. \tag{5}$$

The $P(t)$ vector represents the population of each of the three species at time t. The matrix D is a differential operator containing the interaction between the species. G contains the the growth or natural death rate constants of the animals. With these (2) will have the form:

$$\begin{aligned}\frac{dp_1}{dt} &= g_1 p_1(t) + d_{1,2} p_1(t) p_2(t) + d_{1,3} p_1(t) p_3(t),\\ \frac{dp_2}{dt} &= d_{2,1} p_2(t) p_1(t) + g_2 p_2(t) + d_{2,3} p_2(t) p_3(t),\\ \frac{dp_3}{dt} &= d_{3,1} p_3(t) p_1(t) + d_{3,2} p_3(t) p_2(t) + g_3 p_3(t).\end{aligned} \tag{6}$$

Now this is one of many possible solutions, but this has its weaknesses as well. What can be seen as a major one is the competition inside a species, which could be represented if the trace of matrix D in (4) was not zero. The truth is, it is contained in matrix G in (5) as the natural growth or death rate.

3 Simulation

First, I wanted to test the parameters I saw in the first reference I listed which are:

$$\begin{aligned} P_1 &= 50,\\ P_2 &= 10,\\ P_3 &= 5,\\ g_1 &= 0.25,\\ g_2 &= -0.5,\\ g_3 &= -0.5,\\ d_{12} &= -0.04,\\ d_{13} &= -0.04,\\ d_{21} &= 0.04,\\ d_{23} &= -0.02,\\ d_{31} &= 0.02,\\ d_{32} &= 0.04. \end{aligned} \tag{7}$$

And the result is:

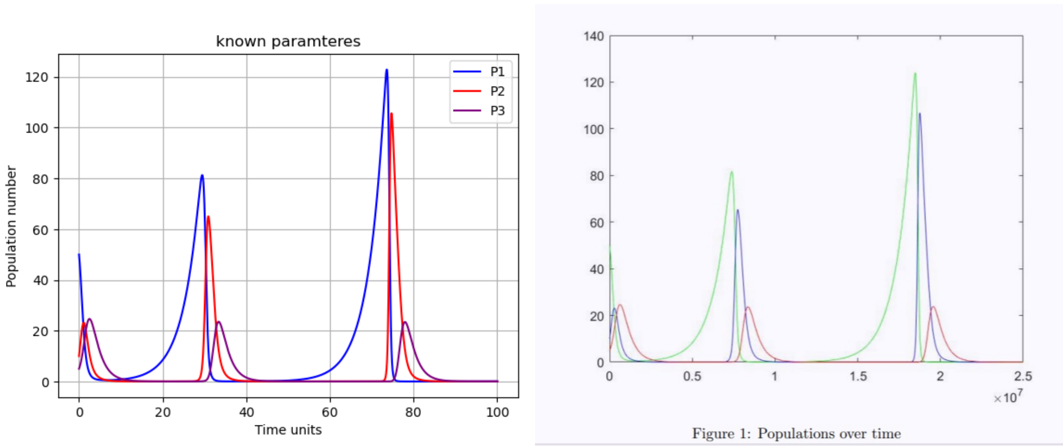

Figure 1: My simulation (left) with parameters I know the result of (right)

The results are the same, or at least so close to each other that they cannot be distinguished. With this I know that my simulation is correct, therefore I can start experimenting with it.

3.1 Experimenting

I wanted to see how does a small ϵ variation effect the result. I chose 3 parameters to change slightly, and plotted them separately. First, I changed g_1. The first figure is made with the known shown in (7), and the rest is generated by adding 0.2 to the previous values.

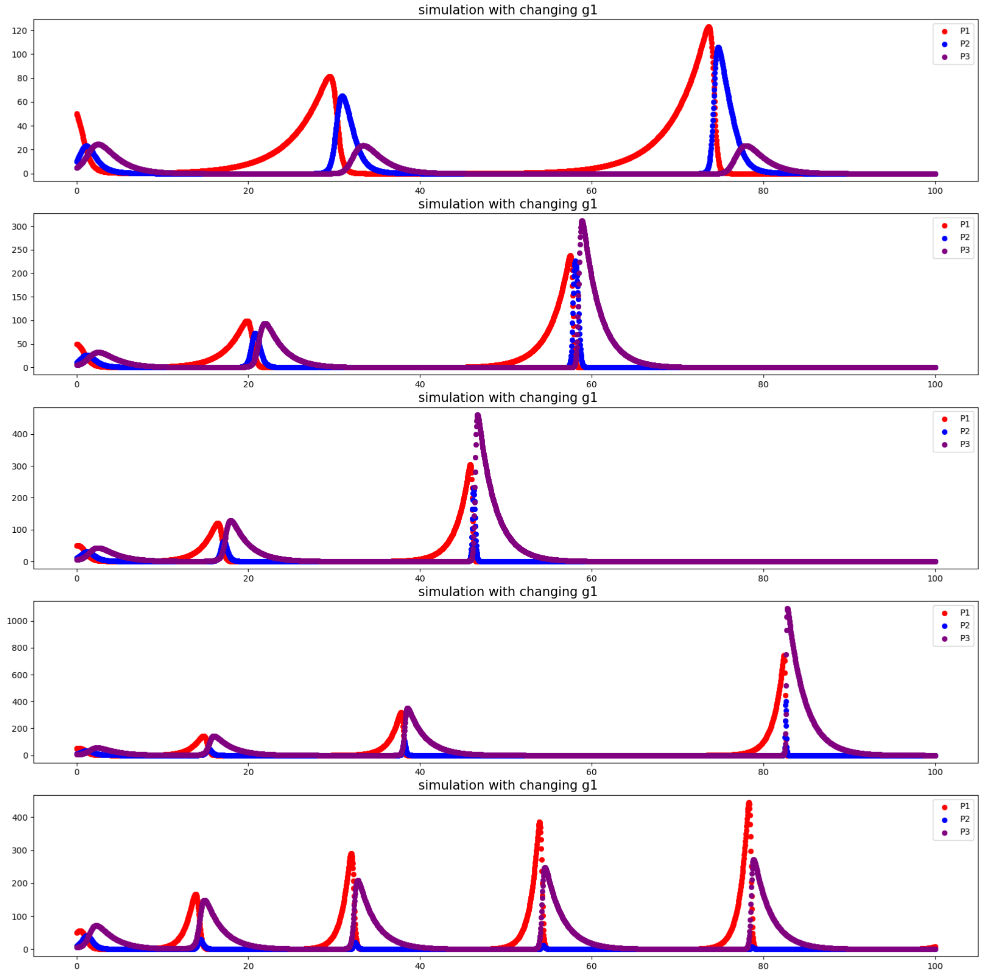

Figure 2: Simulation with changing g_1. The first is the same as on Figure 1. and the rest differs by $g_1 = +0.2$ each time. As we can see, while increasing the natural growth rate should be beneficial to the corresponding P_1, at first it is not. It is beneficial only in the last cycle, where it is clearly more advantageous to the species.

Next, I increased the starting population number P_2 by 10 each time.

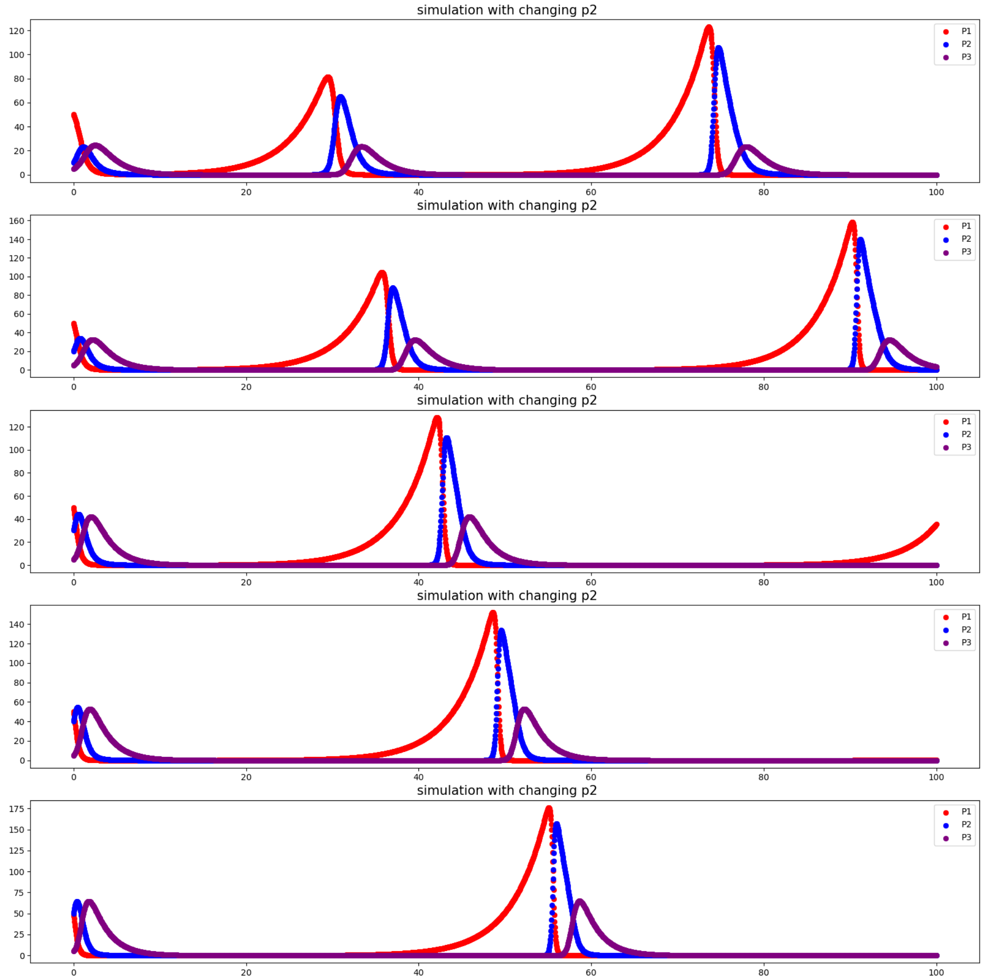

Figure 3: Simulation with changing P_2. The first is the same as on Figure 1. and the rest is different by $P_2 = +10$ each time. Increasing one of the population numbers is not only advantageous for one, but for all the species.

Lastly, I changed d_{12} and d_{21} by 0.03 and -0.03 respectively. I did it this way because, while numerically I can change only one of them, I think changing the interaction between two species from only one side is not life-like.

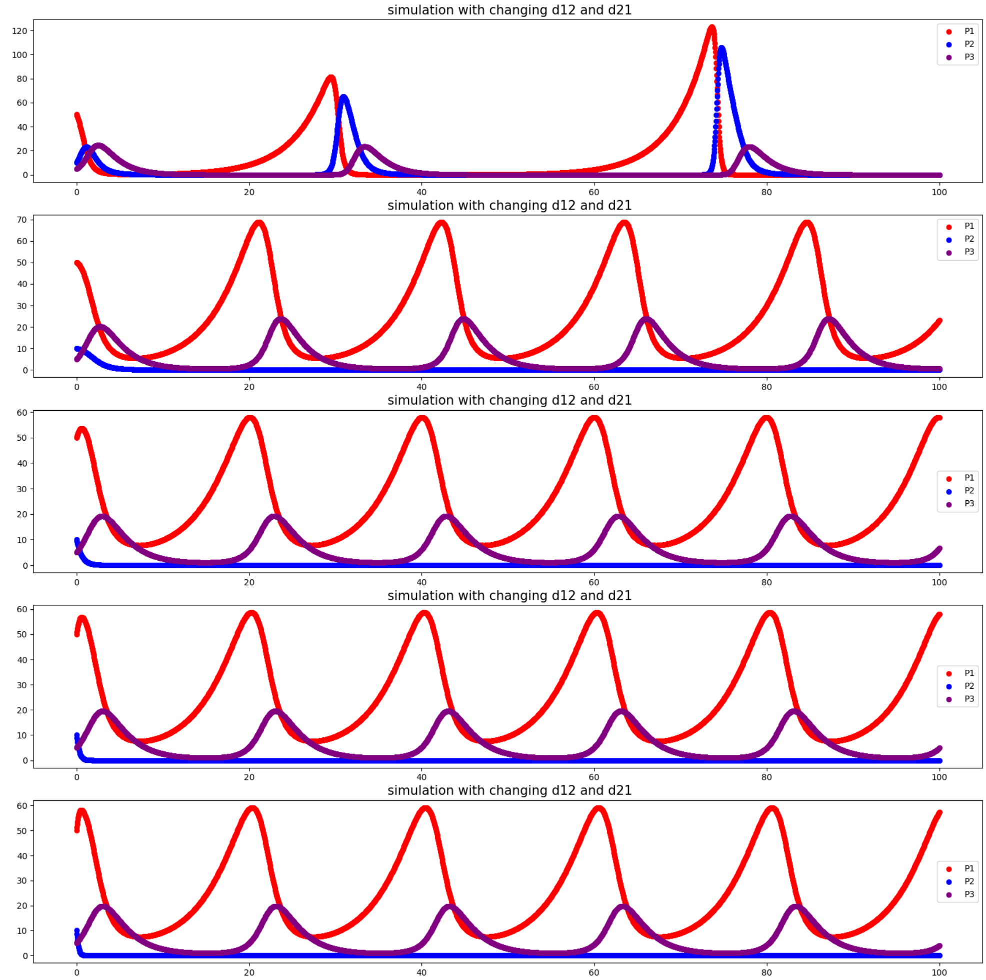

Figure 4: Simulation with changing d_{12} and d_{21}. The first is the same as on Figure (1). and the rest differs by a factor of $d_{12} = +0.03$ and $d_{21} = -0.03$ each time. The system is more sensitive to the change of the interaction parameters. We can see that P_1 starts to dominate the other two really fast.

3.2 Extinction

Next, I wanted a set of parameters, when one of the species goes extinct, and the others live like nothing happened. This was achieved by the following parameters:

$$\begin{aligned} P_1 &= 50, \\ P_2 &= 30, \\ P_3 &= 40, \\ g_1 &= 1.8, \\ g_2 &= -1.6, \\ g_3 &= -2.6, \\ d_{12} &= -0.04, \\ d_{13} &= -0.03, \\ d_{21} &= 0.04, \\ d_{23} &= -0.02, \\ d_{31} &= 0.03, \\ d_{32} &= 0.02. \end{aligned} \tag{8}$$

And the resulting plot is:

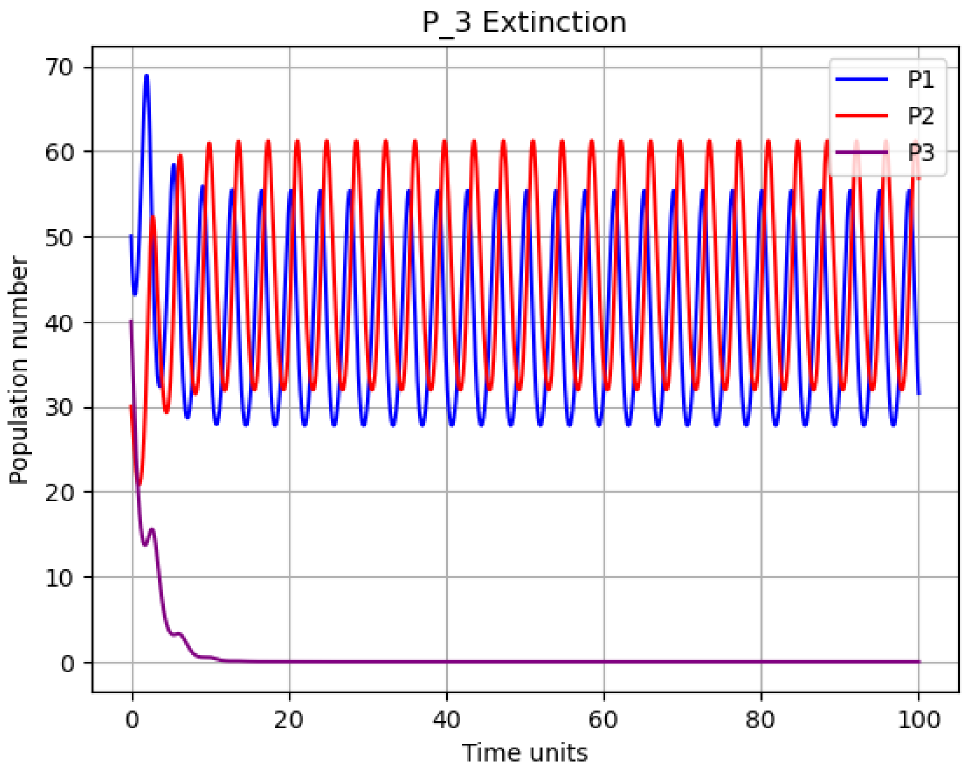

Figure 5: Simulation where P_3 goes extinct. While its effect can be seen, it did not change much for the remaining two.

3.3 Stable population

To get a stable population we need equation (6) to be zero. Let's say $P_1 = 50,\ P_2 = 30,\ P_3 = 20$. Our equation from (6) will be:

$$\begin{aligned} 50 \cdot (g1 + 30 \cdot d_{12} + 20 \cdot d_{13}) &= 0, \\ 30 \cdot (g2 + 50 \cdot d_{21} + 20 \cdot d_{23}) &= 0, \\ 20 \cdot (g3 + 50 \cdot d_{31} + 30 \cdot d_{32}) &= 0. \end{aligned} \tag{9}$$

For these to be equal to zero, the expression inside the brackets needs to be zero. One possible combination of this is:

$$\begin{aligned} g_1 = g2 = g3 &= 0, \\ d_{12} &= 0.04, \\ d_{13} &= -0.06, \\ d_{21} &= -0.04, \\ d_{23} &= 0.1, \\ d_{31} &= 0.03, \\ d_{32} &= -0.05. \end{aligned} \tag{10}$$

The resulting plot is:

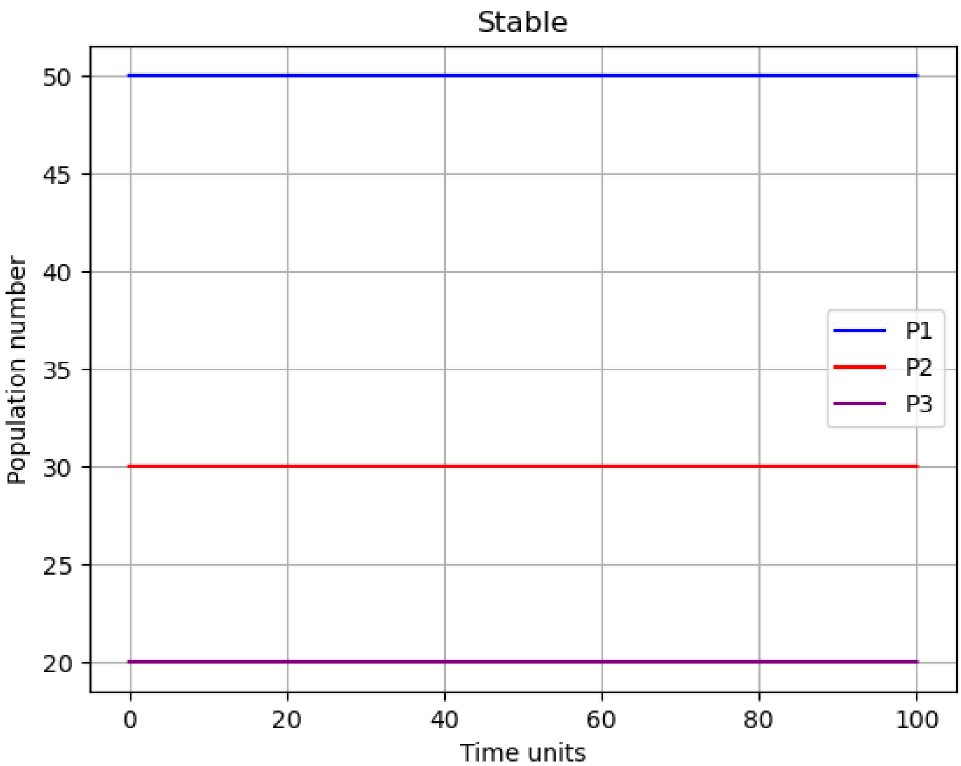

Figure 6: Simulation where all three population numbers remain the same.

3.4 Population numbers relative to each other

For this part I examined the population numbers relatively to each other. What I want the result to be is a circle-like shape witch means their population in cyclical. One possible parameter set is:

$$
\begin{aligned}
P_1 &= 50,\\
P_2 &= 30,\\
P_3 &= 20,\\
g_1 &= 1.8,\\
g_2 &= 0.4,\\
g_3 &= -0.6,\\
d_{12} &= -0.04,\\
d_{13} &= -0.03,\\
d_{21} &= 0.04,\\
d_{23} &= -0.02,\\
d_{31} &= 0.03,\\
d_{32} &= 0.02.
\end{aligned}
\tag{11}
$$

I also wanted to see how stable it is with respect to little change so I made a for loop with $dt = 0.025$ and checked 4 instances before, and 5 after the "circle".

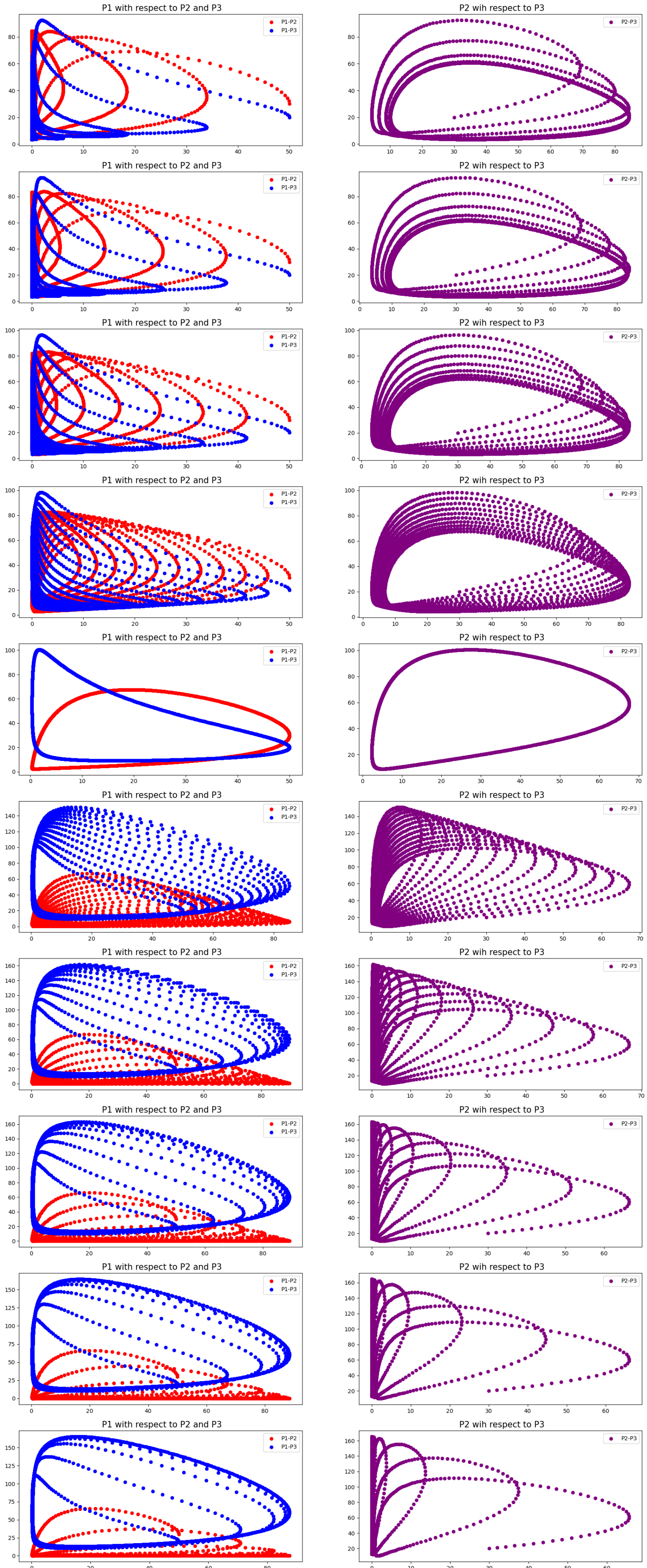

Figure 7: Population numbers with respect to each other. It can be seen, that this equilibrium is quite sensitive to change.

3.5 Mutually bad and good instances

Lastly I wanted to simulate how it would look like if two species could coexist and even benefit from each other, or if they were competition to each other. The parameters for the mutually bad coexistence are the following:

$$
\begin{aligned}
P_1 &= 50, \\
P_2 &= 30, \\
P_3 &= 25, \\
g_1 &= 0.25, \\
g_2 &= 0.2, \\
g_3 &= -0.4, \\
d_{12} &= -0.04, \\
d_{13} &= -0.04, \\
d_{21} &= -0.04, \\
d_{23} &= -0.02825, \\
d_{31} &= 0.02, \\
d_{32} &= 0.04.
\end{aligned} \tag{12}
$$

The result is:

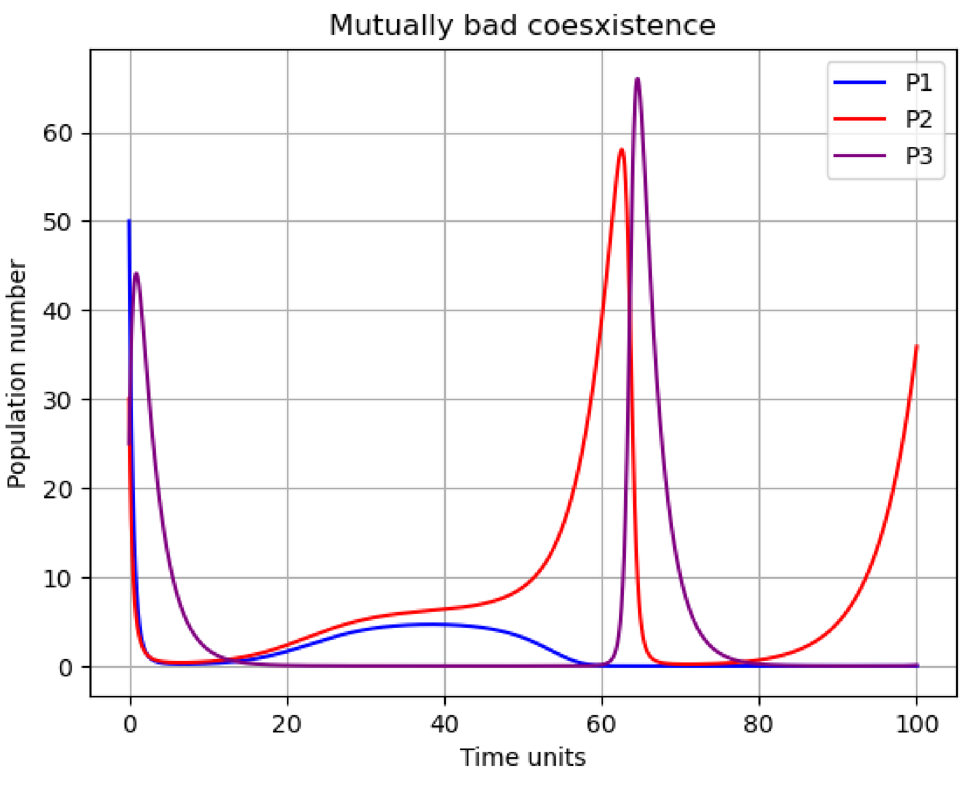

Figure 8: Simulation where two species are detrimental to each other. It can be seen that at first they are both close to extinction, but after P_1 died out there was a spike in P_2's population.

The parameters for the mutually good coexistence are:

$$\begin{aligned}
P_1 &= 50, \\
P_2 &= 40, \\
P_3 &= 25, \\
g_1 &= 0.25, \\
g_2 &= 0.2, \\
g_3 &= 0.04, \\
d_{12} &= 0.04, \\
d_{13} &= 0.04, \\
d_{21} &= 0.04, \\
d_{23} &= 0.02825, \\
d_{31} &= 0.02, \\
d_{32} &= 0.04.
\end{aligned} \tag{13}$$

And the result is:

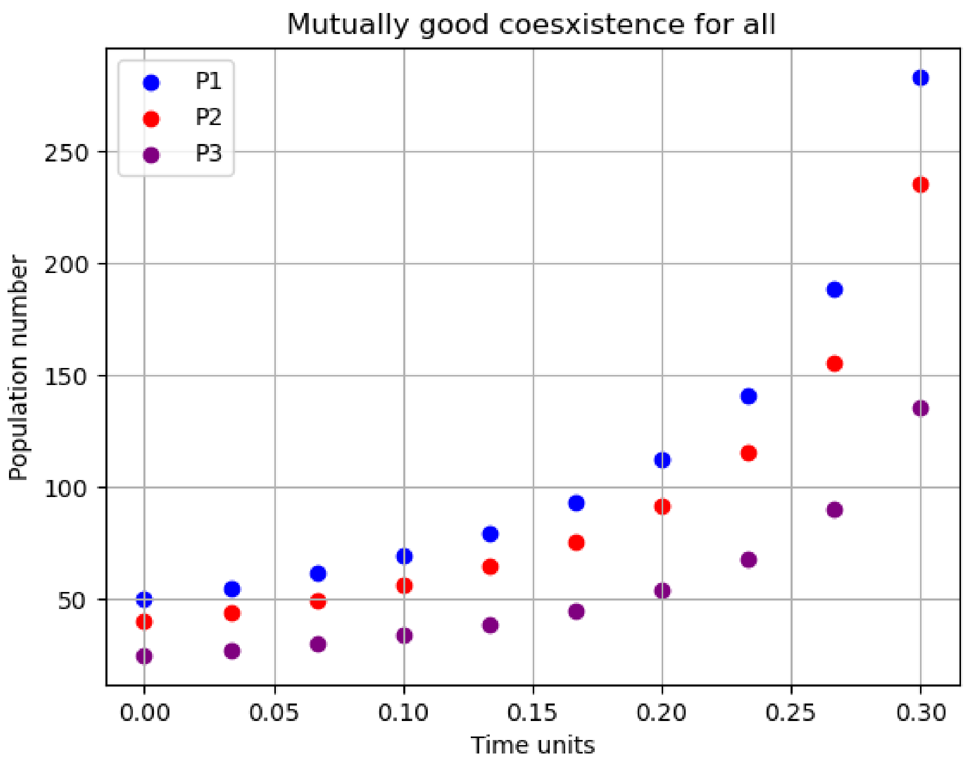

Figure 9: Simulation where coexistence benefits everyone. The population number boomed very fast (it was 10^{128} when I simulated under 100 time units).

4 Discussion

My simulation works the same as it does in the literature shown on Figure (1). It works as expected, as in, I found a parameter set for every special case I wanted to. I simulated cyclic coexistence, experimented with small changes, found parameters where only one species died out, where every population was static, made a plot where the species are beneficial to each other and one where they are detrimental to each other. The one I like the most is where I experimented with the population number of a species with respect to another species' population number (the plots looks good).

5 References

https://sites.math.washington.edu/~morrow/336_18/2016papers/lalith.pdf

http://csabai.web.elte.hu/http/szamszim/simLec6.pdf

https://www.researchgate.net/publication/272396279_Analysis_of_three_species_Lotka-Vol

https://scientific-python.readthedocs.io/en/latest/notebooks_rst/3_Ordinary_Differenti